# Case Study on Cloud Service Models (IaaS, PaaS and SaaS) Explored Through Free-Tier Platforms, with Zoho Creator as the Reference Application Platform

## Aim

To study how Infrastructure as a Service, Platform as a Service and Software as a Service divide responsibility between a cloud provider and its customer, to test what free-tier accounts really give a beginner, and to establish where Zoho Creator belongs in this model for a small business application.

## Objective

- Describe the three service models through the layers of the technology stack that each one hands over to the provider.
- Compare the current free offers of AWS, Google Cloud, Microsoft Azure and Zoho Creator.
- Trace one business requirement through four delivery routes and compare the effort each route demands.
- Classify Zoho Creator using evidence from its hosting, security and licensing documentation.

## Selected Scenario

**Thulliyam Instrument Care (TIC)** is a fictional 14-person firm in Chennai that repairs and calibrates industrial instruments such as pressure gauges, temperature transmitters and flow meters for about sixty recurring customers. Service requests arrive by phone and messaging apps and are written into a shared spreadsheet. TIC wants one Service Request tracker that staff and customers can open from a browser or a phone. It has no full-time IT administrator, a modest budget and a preference for keeping customer data inside India. The same fictional firm appears in the companion case studies of this series. Scenario figures are illustrative assumptions; platform facts come from public vendor documentation.

## 1. Service Models and the Responsibility Stack

Every cloud service model answers one question: which layers of the stack does the customer still have to look after? Moving from on-premises to IaaS, PaaS and SaaS, responsibility steadily passes to the provider, while the customer's freedom to change the system narrows in step. Figure 1 shades the layers a provider manages. Zoho Creator is shown as a fifth column because it does not behave like a plain SaaS product: the customer builds the application, yet never touches anything below the application and data layers.

| Stack layer | On-premises | IaaS | PaaS | SaaS | Zoho Creator (aPaaS) |
|---|---|---|---|---|---|
| Application logic and user interface | **You** | **You** | **You** | Provider | **You** |
| Data and access rules | **You** | **You** | **You** | **You** | **You** |
| Runtime and middleware | **You** | **You** | Provider | Provider | Provider |
| Operating system | **You** | **You** | Provider | Provider | Provider |
| Virtualisation, servers, storage | **You** | Provider | Provider | Provider | Provider |
| Network and data centre | **You** | Provider | Provider | Provider | Provider |

☐ Customer manages ▒ Provider manages

*Figure 1: Who manages each layer under each service model (shaded = provider, white = customer)*

## 2. What the Free Tiers Actually Offer

Free tiers exist for learning, and their terms change quickly. AWS, for example, stopped giving new accounts a fixed twelve months of free usage after 15 July 2025 and moved to a credit-based Free plan. Table 1 records the position in October 2026.

*Table 1: Free-tier offers compared (October 2026)*

| Platform | Starting offer | Time limit | What stays free | Model it exposes |
|---|---|---|---|---|
| AWS | Free plan: USD 100 credit at sign-up and up to USD 100 more for completing exploration tasks | Up to 6 months or until credits run out; the account then closes unless upgraded to the Paid plan | 30+ always-free services within monthly limits (for example Lambda, DynamoDB, CloudFront) | IaaS (EC2 servers) and serverless PaaS (Lambda) |
| Google Cloud | USD 300 welcome credit | 90 days; the trial billing account closes if credit is spent or time lapses without an upgrade | One e2-micro VM per month in eligible US regions, 5 GB-months of standard storage and other products | IaaS (Compute Engine) and managed PaaS products |
| Microsoft Azure | USD 200 credit | 30 days for the credit | Selected services free for 12 months, plus a few always-free services | IaaS (virtual machines) and PaaS services |
| Zoho Creator | 15-day full-featured trial of the Professional plan, no credit card | After 15 days the account moves to the Free Edition | Free Edition supports one custom application | aPaaS delivered as a service (Section 4) |

Azure figures come from independent 2026 summaries of Microsoft's free-account terms and should be rechecked at sign-up. The pattern matters more than the numbers: none of the three public clouds provides a lasting free production environment, because credits end between 30 days and 6 months and only small always-free allowances persist. Free tiers therefore suit experiments and proofs of concept, not the daily running of TIC's tracker.

## 3. One Requirement, Four Delivery Routes

To compare effort fairly, the same requirement (an online Service Request tracker with forms, approvals and reports) is delivered four ways. Table 2 shows what TIC would have to build and keep doing.

*Table 2: One tracker, four routes (ratings are the author's qualitative judgement)*

| Route | What TIC builds or configures | Skills needed | Ongoing duties | Flexibility | Fit for TIC |
|---|---|---|---|---|---|
| A. IaaS virtual machine on a free-tier cloud | Server, operating system, web server, database, application code, login and backup scripts | System administration, web development, security hardening | Patching, backups, certificates, monitoring, scaling | Very high | Poor: operating burden is out of proportion to the need |
| B. PaaS managed code platform | Application code and database schema deployed to a managed runtime | Software development | Code maintenance, dependency updates, access control | High | Fair: still needs a developer |
| C. SaaS ready-made ticketing tool | Field and user configuration only | Administration | User management, data exports | Low to medium | Fair: quick start, but calibration-cycle logic is hard to model |
| D. aPaaS Zoho Creator | Forms, workflows, reports and pages through visual builders and Deluge scripts | Business analysis, light scripting | Access permissions, data quality, application changes | Medium to high | Good: custom fit without infrastructure work |

## 4. Where Zoho Creator Sits

**Classification.** Zoho describes Creator as a low-code application development platform. Analysts group such products as enterprise low-code application platforms; Gartner used the title "high-productivity application platform as a service" for the same market in 2017 and 2018. The older name explains the hybrid nature: users reach Creator through a browser like SaaS, but they use it to build applications like PaaS.

**Hosting.** Creator runs on Zoho's own infrastructure rather than on a public cloud. Zoho's infrastructure FAQ states that a co-location provider looks after the building, power, cooling and physical security while Zoho supplies the servers and storage, and that application data is replicated across data centres. Zoho documents AES-256 encryption at rest, TLS in transit and certifications that include ISO 27001, ISO 27017, ISO 27018 and SOC 2 Type II.

**Residency.** Each Zoho data centre holds only the data of accounts registered on its own domain, and the data centre is chosen automatically from the sign-up location. An account opened from India is therefore served from the India data centre, which meets TIC's preference without any server decision.

**What TIC still owns.** Responsibility does not disappear. TIC remains accountable for who may see which records, the quality of the data entered, the logic of its workflows and the cost of user licences, which Zoho charges per user.

## 5. Exploration Log and Cost Guardrails

The lab value of a free tier lies in doing one small task on each model and recording what the provider handled for you. Table 3 lists a suitable activity for each model together with the evidence worth capturing.

*Table 3: A one-task-per-model exploration plan*

| Model | Free-tier activity | What it teaches | Evidence to capture |
|---|---|---|---|
| IaaS | Launch the smallest virtual machine offered, connect to it remotely and serve a test web page | The customer owns the operating system, firewall rules and updates | Instance dashboard and the served page |
| PaaS | Deploy a small function or web application to a managed runtime | The provider runs, patches and scales the runtime; the customer ships code | Deployment log and public address |
| SaaS | Open an account in a ready-made web tool and configure users and fields | Only configuration is possible; the product cannot be restructured | Settings screens |
| aPaaS | Create a Zoho Creator application on the Free Edition and add one form | Applications can be built without servers, yet remain inside the vendor's rules | Form, default report and application link |

Two guardrails apply to every public-cloud account. First, read the usage page that shows free allowances against credits: AWS, for instance, sends notices when remaining credit falls to 50, 25 and 10 per cent. Second, remove every resource when the exercise ends, because on the AWS Free plan the account itself closes when the credit or the six months run out, and the data is held only for a limited period before deletion.

## Results

- IaaS, PaaS and SaaS differ chiefly in how many layers the customer must still operate; Zoho Creator hands over everything below the application and data layers.
- Free tiers teach the models well but expire within 30 days to 6 months, so they cannot host a lasting production tracker.
- Of the four routes, Zoho Creator gave the best balance of custom fit and low operating burden for a firm with no IT staff.
- The India data centre satisfies the residency preference without extra configuration.

## Conclusion

Zoho Creator is best classified as an application platform as a service consumed in the manner of SaaS. Choosing it moves TIC's effort from running servers to designing processes. The price of that convenience is dependence on a single vendor and its proprietary Deluge language, a trade-off TIC should accept deliberately rather than discover later.

## Sources Consulted


- AWS Free Tier page (aws.amazon.com/free) and AWS Builder Center free-tier guides, 2026
- Google Cloud free trial and Free Tier documentation (cloud.google.com/free)
- Independent free-tier summaries: freetier.co and itechguides.com, 2026 (Azure terms)
- Zoho Creator pricing page and FAQs, including the infrastructure, privacy and security FAQs (zoho.com, help.zoho.com)
- Zoho Creator product catalogue (security and compliance), help.zoho.com
- OutSystems summary of the Gartner Magic Quadrant for Enterprise Low-Code Application Platforms (market naming history)


*Platform facts were checked against public vendor documentation in October 2026; all text is written in the author's own words.*